Kirill Korolev

Research Intern

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Personal Profile

M.Sc. in Mathematics of Machine Learning and B.Sc. in Applied Mathematics and Computer Science from HSE University, Moscow. Machine learning researcher working at the intersection of generative modeling, reinforcement learning, and probabilistic inference. My research focuses on developing scalable algorithms for sampling, generative modeling, with recent work on entropy-regularized reinforcement learning and diffusion-based samplers.

I am broadly interested in generative models, reinforcement learning, diffusion models, Monte Carlo methods. My work combines theoretical analysis with practical algorithm design.

Education

Higher School of Economics — *Moscow, Russia*

Masters Degree "Math of Machine Learning" — *Sept 2024 - June 2026*

- GPA 9.5

Higher School of Economics — *Moscow, Russia*

Bachelor Degree in Applied Math and Computer Science — *Sept 2020 - June 2024*

- GPA 8.27

Yandex School of Data Analysis — *Moscow, Russia*

Sept 2022 - June 2024

Work Experience

HDI Lab — *Moscow, Russia*

Research intern under supervision of **Sergey Samsonov** — *Mar 2025 – Present*

- Research on diffusion-based samplers, adaptive MCMC, RL, acceleration of diffusion models using RL finetune.

Glam AI — *Moscow, Russia*

Computer vision engineer — *Aug 2024 – Mar 2025*

- Developed diffusion-based virtual try-on systems and personalization pipelines.

Yandex Research — *Moscow, Russia*

DL engineer & researcher — *Jun 2023 – Jun 2024*

- Conducted research on diffusion models and large-scale retrieval systems.
- Developed transformer-based retrieval, ranking, and LLM-powered search components in geo search.

Huawei Russian Research Institute — *Moscow, Russia*

DL engineer — *Nov 2022 – May 2023*

- Applied computer vision and model compression techniques to video processing systems.

Higher School of Economics — *Moscow, Russia*

Seminar tutor — *Sep 2021 – Dec 2025*

- Taught seminars in Deep Learning, Linear Algebra, and Statistics.

Publications & Projects

Submissions at ICLR 2027 and AISTATS 2027 — *October 2026*

- Under review.

Numerical Integration on Bounded Domains via Learned Reflected Diffusions — *September 2026*

- Oral at NeurIPS 2026 STODY Workshop. **Paper**, **Source code**.

Stop the Sampler! Classifier-Based Adaptive Stopping for Sampling Kernels — *May 2026*

- Accepted to NeurIPS 2026 Main Track and ICML 2026 SPIGM Workshop. **Paper**, **Source code**.

Thesis "Development of domain adaptation methods for generative models" — *May 2024*

- An encoder based approach for adapting a GAN for a new domain. **Source code**.

Achievements

2024 **2nd Degree Diploma**, Skoltech-HSE Olympiad Math of Machine Learning — *Moscow, Russia*